

Create and Execute Attendance Processor DAG

1. Run Apache Airflow Container

```
docker run -d -p 8080:8080 --name airflow apache/airflow standalone
```

2. Verify Running Container

```
docker ps
```

3. Retrieve Login Credentials

```
docker logs airflow
```

4. Access the Container

```
docker exec -it airflow bash
```

5. Navigate to DAGs Folder

```
cd /opt/airflow/dags
```

6. Create DAG File

```
cat > attendance_dag.py
```

Paste the code

```
from airflow import DAG  
from airflow.operators.python import PythonOperator  
from datetime import datetime
```

```
def create_attendance_file():  
    with open("/tmp/attendance.txt", "w") as f:  
        f.write("Aarav,Present\n")  
        f.write("Priya,Present\n")  
        f.write("Rahul,Absent\n")  
        f.write("Sneha,Present\n")  
        f.write("Kiran,Absent\n")  
        f.write("Ananya,Present\n")  
        f.write("Vikram,Present\n")  
        f.write("Meera,Absent\n")  
        f.write("Farhan,Present\n")
```

```
f.write("Divya,Present\n")
```

```
def read_attendance_file():  
    with open("/tmp/attendance.txt", "r") as f:  
        for line in f:  
            print(line.strip())
```

```
def count_total_students():  
    count = 0  
    with open("/tmp/attendance.txt", "r") as f:  
        for _ in f:  
            count += 1  
    print(f"Total Students = {count}")  
    return count
```

```
def count_present_students():  
    present = 0  
    with open("/tmp/attendance.txt", "r") as f:  
        for line in f:  
            if "Present" in line:  
                present += 1  
    print(f"Present Students = {present}")  
    return present
```

```
def count_absent_students():  
    absent = 0  
    with open("/tmp/attendance.txt", "r") as f:  
        for line in f:  
            if "Absent" in line:  
                absent += 1  
    print(f"Absent Students = {absent}")  
    return absent
```

```
def calculate_attendance_percentage(ti):  
    total = ti.xcom_pull(task_ids="count_total_students")  
    present = ti.xcom_pull(task_ids="count_present_students")  
    percentage = (present / total) * 100  
    print(f"Attendance Percentage = {percentage}%")  
    ti.xcom_push(key="percentage", value=percentage)
```

```

def list_absent_students():
    print("Absent Students List:")
    with open("/tmp/attendance.txt", "r") as f:
        for line in f:
            name, status = line.strip().split(",")
            if status == "Absent":
                print(name)

def generate_attendance_report(ti):
    total = ti.xcom_pull(task_ids="count_total_students")
    present = ti.xcom_pull(task_ids="count_present_students")
    absent = ti.xcom_pull(task_ids="count_absent_students")
    percentage =
ti.xcom_pull(task_ids="calculate_attendance_percentage",
key="percentage")

    status = "Good" if percentage >= 75 else "Needs Improvement"

    with open("/tmp/attendance_report.txt", "w") as f:
        f.write("Daily Attendance Report\n")
        f.write(f"Total Students = {total}\n")
        f.write(f"Present Students = {present}\n")
        f.write(f"Absent Students = {absent}\n")
        f.write(f"Attendance Percentage = {percentage}%\n")
        f.write(f"Status = {status}\n")

with DAG(
    dag_id="attendance_processor_dag",
    start_date=datetime(2025, 1, 1),
    schedule="@daily",
    catchup=False
) as dag:

    t1 = PythonOperator(task_id="create_attendance_file",
python_callable=create_attendance_file)
    t2 = PythonOperator(task_id="read_attendance_file",
python_callable=read_attendance_file)
    t3 = PythonOperator(task_id="count_total_students",
python_callable=count_total_students)
    t4 = PythonOperator(task_id="count_present_students",
python_callable=count_present_students)
    t5 = PythonOperator(task_id="count_absent_students",
python_callable=count_absent_students)
    t6 =

```

```

PythonOperator(task_id="calculate_attendance_percentage",
python_callable=calculate_attendance_percentage)
    t7 = PythonOperator(task_id="list_absent_students",
python_callable=list_absent_students)
    t8 = PythonOperator(task_id="generate_attendance_report",
python_callable=generate_attendance_report)

```

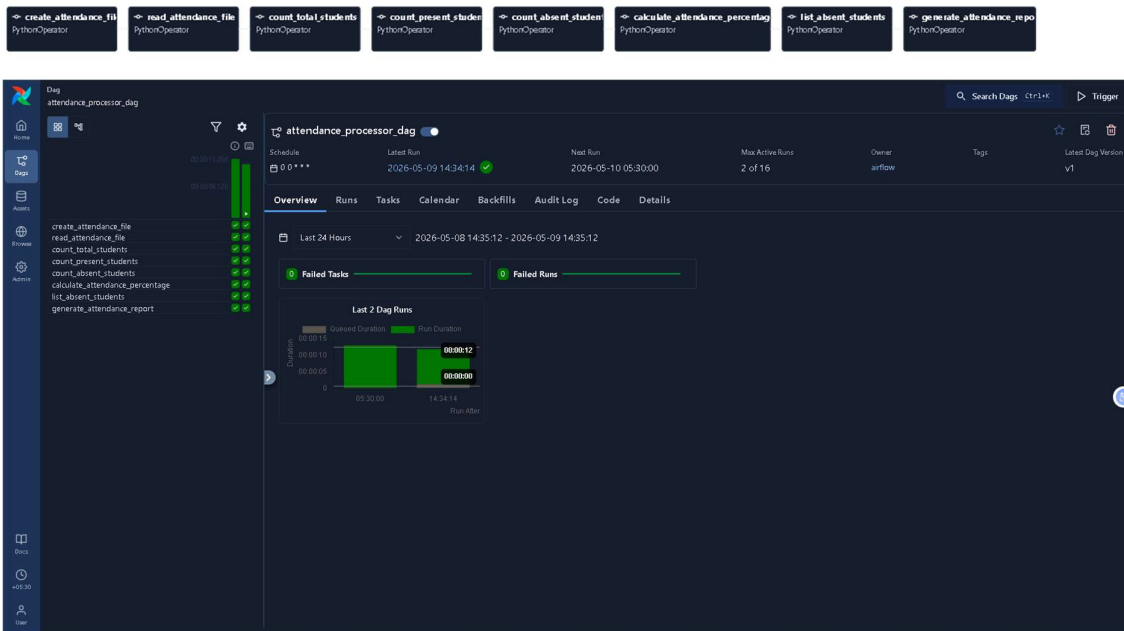
```
t1 >> t2 >> t3 >> t4 >> t5 >> t6 >> t7 >> t8
```

7. Verify File Creation

```
ls
```

8. Exit Container

```
exit
```



```

root@ubuntu:~$ docker exec -it airflow bash
airflow@3eb2eb237a40:/opt/airflow$ cat /tmp/attendance_report.txt
Daily Attendance Report
Total Students = 10
Present Students = 7
Absent Students = 3
Attendance Percentage = 70%
Status = Needs Improvement
airflow@3eb2eb237a40:/opt/airflow$

```